IPL ANALYSIS

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Course:
DATA ANALYTICS

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MODULES

- POWERBI – For Building Dashboard

Introduction

This project analyzes the **Indian Premier League (IPL)** data from multiple seasons to derive key insights about player performance, team success, and match trends.

- **Matches table:** Contains match-level data (venue, date, teams, winner, etc.).
- **Deliveries table:** Contains ball-by-ball data (batsman, bowler, runs, dismissal types, etc.).

These tables are connected in a **one-to-many relationship** (Matches → Deliveries) on the id field.

The goal of this dashboard is to present an interactive, slicer-driven analysis where users can filter by **team** or **season** to view performance trends, top players, and match outcomes.

Methodology

Data Source & Cleaning:

- Data was collected from historical IPL datasets (2008–2017).

- Null values were handled (e.g., missing cities).
- Columns were standardized for consistency.

Data Modeling in Power BI:

- Tables: matches and deliveries connected via id.
- Measures created using **DAX** for KPIs: Total Runs, Wickets, Centuries, Half-Centuries, etc.
- Relationships set to ensure season-level and team-level filtering.

DAX Measures Used (Examples):

-- Total Runs

Total Runs = SUM(deliveries[batsman_runs]) + SUM(deliveries[extra_runs])

-- Total Wickets

Total Wickets = COUNTROWS(FILTER(deliveries, NOT ISBLANK(deliveries[dismissal_kind])))

-- Total Fours

Total Fours = CALCULATE(COUNTROWS(deliveries), deliveries[batsman_runs] = 4)

-- Total Sixes

Total Sixes = CALCULATE(COUNTROWS(deliveries), deliveries[batsman_runs] = 6)

-- Total Matches

Total Matches = COUNTROWS(matches)

-- Total Teams

Total Teams = DISTINCTCOUNT(matches[team1]) + DISTINCTCOUNT(matches[team2])

-- Centuries

Centuries = COUNTROWS(

```
    FILTER(
        SUMMARIZE(deliveries, deliveries[match_id], deliveries[batsman],
"Runs", SUM(deliveries[batsman_runs])),
        [Runs] >= 100
    )
)

-- Half-Centuries
HalfCenturies = COUNTROWS(
    FILTER(
        SUMMARIZE(deliveries, deliveries[match_id], deliveries[batsman],
"Runs", SUM(deliveries[batsman_runs])),
        [Runs] >= 50 && [Runs] < 100
    )
)
```

3. Key Findings and Analysis

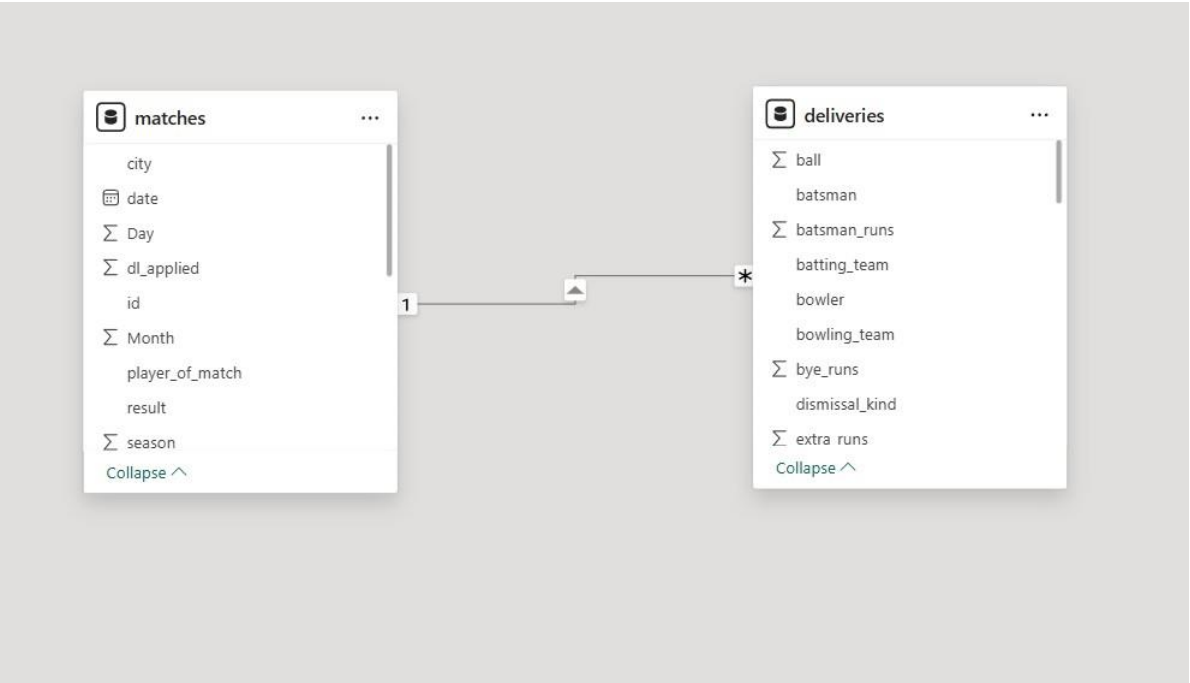
IPL Season Overview

1. **Batsman Runs (Top Run-Scorers)**
   - Horizontal bar chart ranking batsmen by total runs.
   - Helps identify most consistent scorers in a given season.
   - Dynamic with slicers for season and batting_team.
2. **Wickets (Top Wicket-Takers)**
   - Bar chart of bowlers with the highest dismissals.
   - Uses dismissal_kind count (ignores blanks).
3. **Player of the Match Awards**
   - Highlights impactful players who frequently influenced match outcomes.
4. **Wins by City**
   - Displays city-wise distribution of match winners.
   - Useful for identifying city-level home ground advantages.
5. **Sum of Total Runs by Year (Line + Area Chart)**
   - Trend analysis of overall scoring levels per season.
   - Shows increase/decrease in batting dominance over the years.
   - Includes a trendline for better visual insight.
6. **Wins by Team**
   - Visualizes which teams dominated a selected season.

IPL Insights & Key Metrics

1. **Season Final Winner and Runner-Up Cards**
   - Displays the final match outcome for the selected season.
2. **Total Sixes & Total Fours**
   - KPI cards showing season totals.
   - Helps measure batting aggression across seasons.
3. **Total Matches and Total Teams**
   - Counts matches played and teams participated.
4. **Centuries and Half-Centuries**
   - Summarizes major batting achievements.
5. **Orange Cap (Top Run-Scorer)**
   - Player with most runs, total runs, and representing team.
6. **Purple Cap (Top Wicket-Taker)**
   - Bowler with most wickets and their team.
7. **Toss Decision Pie Chart**
   - Shows percentage of toss-winning teams choosing to bat or field.
   - Provides tactical trends for teams across seasons.

4. Final Dashboard



IPL ANALYSIS

Mumbai Indians
Season Final Winner

Chennai Super Kings
Season Final RunnerUp

season: 2015

Total Sixes	Total Fours	Total Matches	Total Teams	Century	Half century
692	2K	59	8	4	89

OrangeCap_Player
DA Warner

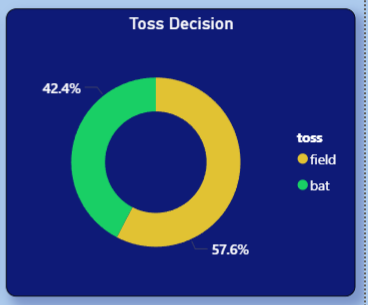
OrangeCap_Runs
562

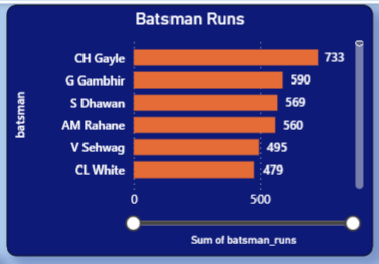
OrangeCap_Team
Sunrisers Hyderabad

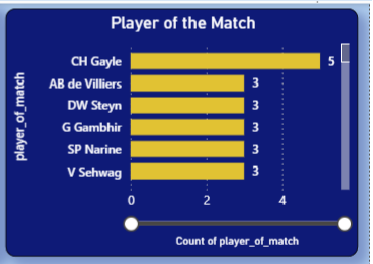
PurpleCap_Player
DJ Bravo

PurpleCap_Wkts
26

PurpleCap_Team
Chennai Super Kings

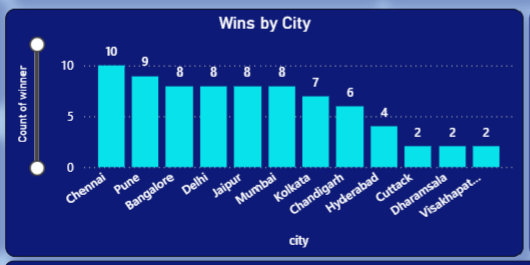


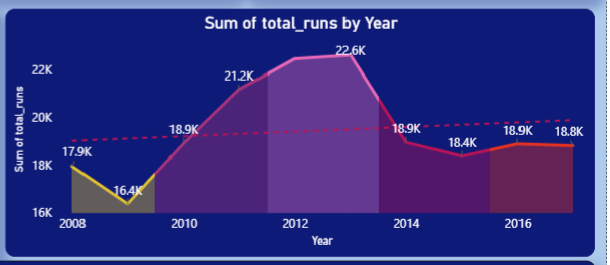


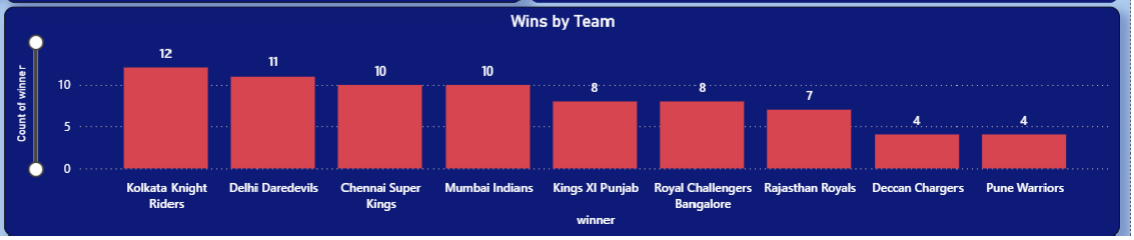


batting_team: All

season: 2012







5. Conclusion

This IPL Analysis Dashboard provides a **comprehensive season-wise and player-wise breakdown** of IPL performance trends. With **interactive slicers**, users can explore:

- How batting and bowling performances evolved seasonally.
- The most impactful players (Orange and Purple Cap holders).
- City-level home advantages and toss decision strategies.

Key insights include:

- Variability in scoring trends across seasons, peaking around 2012–2013.
- Clear identification of dominant players and teams.
- Tactical patterns (bat vs. field choices) for toss-winning teams.

This dashboard can serve **analysts, cricket boards, franchises, and fans** as a decision-support tool to understand historical performance and strategize future IPL campaigns.